

BUSINESS AND PROJECT MANAGEMENT — UNIVERSITÀ DI PISA

Master's Degree in Artificial Intelligence and Data Engineering

AlmReady

An AI-powered virtual exam panel for oral-exam preparation

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SUMMARY

What we'll cover

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Problem & AI-Enabled Solution

Why oral-exam anxiety is real and how AlmReady's multi-agent panel solves it

02

Competition & Positioning

Direct competitors and our defensibility

03

Business Model & Pricing

Business Model Canvas, AI cost structure and freemium tiers

04

Market Sizing & Financials

TAM/SAM/SOM, Year-1 economics and break-even analysis

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Personas & Validation

Who we build for and how we test our riskiest assumptions

06

Risk & Failure Analysis

Negative backcasting: what could kill AlmReady and how we watch for it

Oral-exam anxiety is the norm, not the exception

Students underperform in oral exams due to performance anxiety and the absence of objective feedback. Current methods (mirror practice, being quizzed by family) give no measurable signal and no real exam pressure.

- Fear of public speaking is the most commonly feared situation in the general population. One third of individuals experience significant public-speaking anxiety
- A study with 42 students, reported that public-speaking anxiety really hurts both grades and future job chances. To help students, teacher used generic AI apps
- A structured practice program with sessions and feedbacks lowered self-reported anxiety by 33.2%

HEALTH-SCIENCE FIRST-YEAR STUDENTS (n=210)

74.8%

moderate stress
before exams

14.3%

high stress
before exams

89%

of students report moderate-to-high pre-exam stress

Three independent evidence streams

A

Academic Search

Iterative Scopus queries
(public-speaking anxiety, multimodal
analysis, agent-based models, edge AI
+ privacy): 6 core queries, refined with
AND/OR/NOT.

P

Patent & Freedom-to-Operate

Lens.org / Espacenet, CPC G09B19/04.
Narrowed 3,688 → 244 → 14 patents.
Identified 3 US patents (territorial only)
and 1 pending EU application
(Samsung). Nothing blocks us.

R

Open Community Sources

r/Universitaly screening, query "ansia
esame orale": ~160 posts, 33 explicitly
about oral-exam anxiety. This confirms
that oral-exam anxiety emerges as a
spontaneously discussed pain point
among italian university students,

*"Letteralmente le cose le so, solo che
mi impanico e mando tutto all'aria"
"Gli esami orali mi distruggono"*

A virtual exam panel, built as a multi-agent system

Following Park et al. (2023), our agents observe, reflect and plan → repurposed from open-ended simulated life into three focused examiner personas.

1

Lead Examiner

GPT-4o-mini

Asks the main subject-matter questions in a calm, structured way. Drives the overall flow.

2

Assistant Examiner

Llama 3.1 8B

Faster-paced, jumps in with more direct follow-up questions and pushes harder at set points.

3

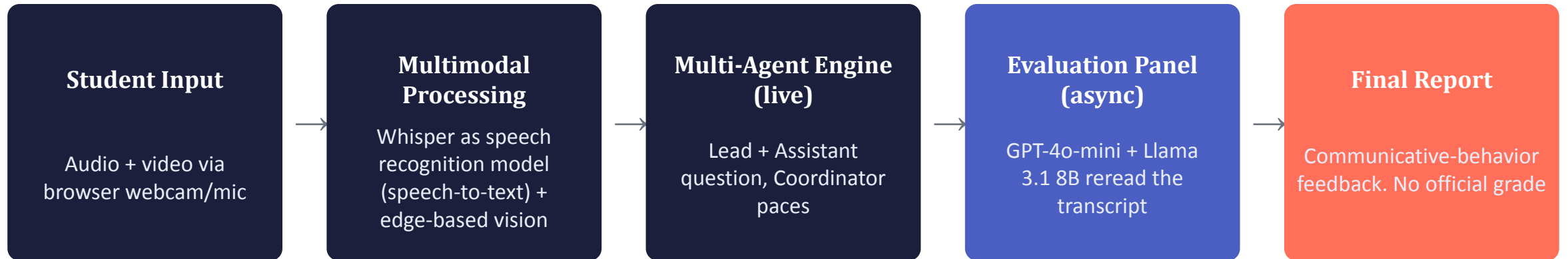
Silent Coordinator

Claude Haiku

Never speaks, works as a controller. Reads transcript and delivery metrics (pauses, speech rate, fillers) and applies pacing rules. Coordinates the two agents.

The coordinator never labels emotional state but only measurable communication metrics, staying within the AI Act's role-play/training exemption.

AlmReady's System Architecture



SECTION 1 · LIVE EXAM PHASE

SECTION 2 · POST-EXAM PHASE

Students upload their own course material so that the examiner agents draw questions from that content, not generic subject questions.

Report covers: content accuracy - clarity of the explanations - speech delivery indicators (pacing, fillers, vocal steadiness) - non-verbal signals (gaze, gesture stability). For each of these dimensions, the report provides a concrete, actionable suggestion. The feedback is always framed as communicative performance, never as an inference about how the student felt.

Two segments, one differentiated value proposition



B2C — Individual Students

Exam Pass, freemium

- Repeatable, objective behavior metrics
- Real pressure from a simulated panel, not a passive listener
- A fraction of the cost of a human tutor



B2B — Universities

Per-student institutional license

- Fewer failed oral exams
- Fewer students falling behind
- Improves overall quality of student services

We are not entering an empty market

Generic AI coaches	Yoodli, Poised	Public speaking / meetings, not exam-specific, no subject panel
Prior-art benchmark	NYU case (Viva)	Voice examiner + multi-LLM grading council: proves the mechanism
VR trainers	PAVE, LLM-VR sims, chatbot Q&A	Reduce speaking anxiety but need dedicated VR hardware
Non-AI substitutes	Tutoring, mirror practice, therapy	Tutoring is not scalable and has high costs. Mirror practice doesn't provide an objective feedback. Therapy has high costs.

Three things nobody else combines at once:
Objective, repeatable behavior metrics · Real simulated exam pressure · A cost structure that scales to many sessions

Porter's Five Forces + Complements

New entrants

HIGH

No patent barrier since the core mechanism is public (NYU case)

Buyer power

MIXED

Students: near-zero switching cost.
Universities: high upfront, lower once onboarded

Substitutes

HIGH

Free alternatives (mirror, family) are the default students already use

Supplier power

HIGH

ElevenLabs voice $\approx$ 90% of marginal session cost

Rivalry

MED-HIGH

Yoodli/Poised have users & funding to add an 'exam mode' fast

Complements

HIGH

OpenAI, Groq, Anthropic, ElevenLabs: today's infrastructure partners, tomorrow's possible rivals

Our defensibility

1

PRIMARY BARRIER

Proprietary interaction dataset

Built from consented real exam-practice sessions. It compounds over time and cannot be replicated without the same user base.

2

SECONDARY BARRIER

Purpose-built specialization

A product built specifically around oral-exam anxiety and subject-aware interrogation, rather than generic public-speaking coaching.



Year 1 is a window of maximum exposure

Neither barrier exists on day one: the dataset only becomes an advantage as usage accumulates and specialization alone can be imitated. In Year 1 our defense is not a barrier but a head start. This is a classic cold start problem: below the Allee critical mass, a data-driven network effect creates no value. Our free tier exists precisely to reach that critical mass fast (freemium as a cold-start tactic) and lock in the design-partner university before any imitator can.

Four Actions Framework (ERRC Grid)



Eliminate

Dedicated hardware and the configuration required by VR-based competitors



Reduce

Costs of hardware and of a human tutor, by relying only on a browser and webcam



Raise

Question specificity, generated from the student's own material, and the realism of simulated exam pressure via a multi-agent panel



Create

An objective, repeatable behavioral feedback tool that is explicitly AI Act-compliant and non-certifying

Business Model Canvas



Local Data Architecture, Built for Privacy

Proprietary Dataset

Initial model training is made on TED-LIUM and a 2026 multimodal oral-presentation dataset. Grows with every consented session: free tier adds transcripts, paid tier adds behavioral metrics.

Edge Computing

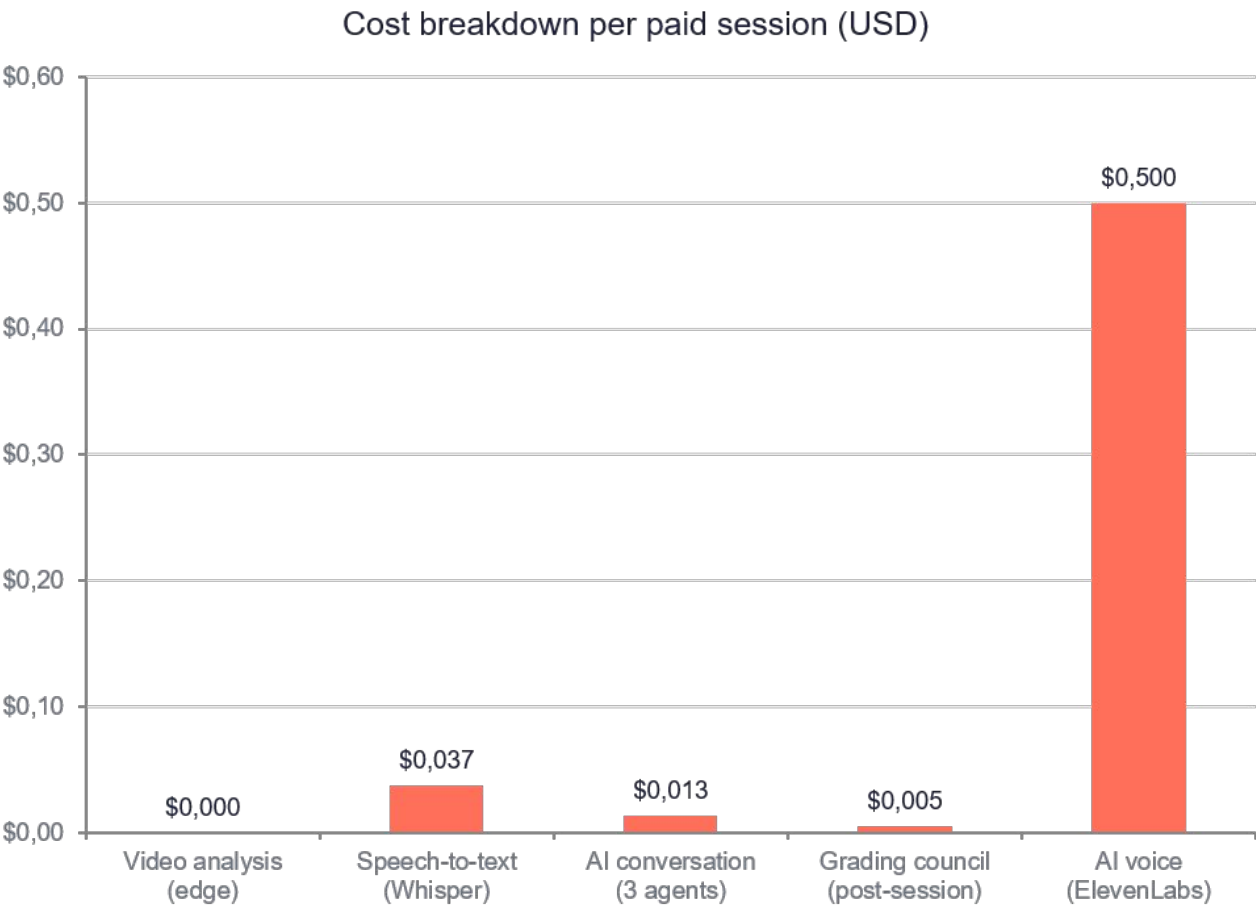
All facial-expression processing runs client-side (MediaPipe / TensorFlow.js). Only lightweight scalar values reach the backend (never raw video or facial imagery).

AI Act Compliance

By never transmitting raw biometric data, the system stays inside the EU AI Act's exemption for role-play and training simulations (Art. 5(1)(f)).

The system only saves and learns from extracted numerical scores. The output is always framed as "communicative-behavior feedback," never as an inferred emotional state.

\$0.55 to run a full 30-minute paid session



Voice generation alone ≈ 90% of marginal cost: our biggest supplier-power risk.

Pricing

F	Free ≤20 sessions/yr Browser TTS, no video	€0
E	Exam Pass 5 full sessions ~65–70% margin	€8–9
I	Institutional per student / year B2B license	~€3

Bottom-up market sizing, B2C and B2B

B2C - REVENUE PER STUDENT (€16 ARPU)

TAM

19.1M students (Europe)

~€306M/yr

SAM

1.4M students (Italy, target fields)

~€22.4M/yr

SOM

~600 paying students (Yr 1)

~€9,600/yr

B2B - REVENUE PER STUDENT (€3/YR LICENSE)

TAM

19.1M students (Europe)

~€57.3M/yr

SAM

1.4M students (Italy, target fields)

~€4.2M/yr

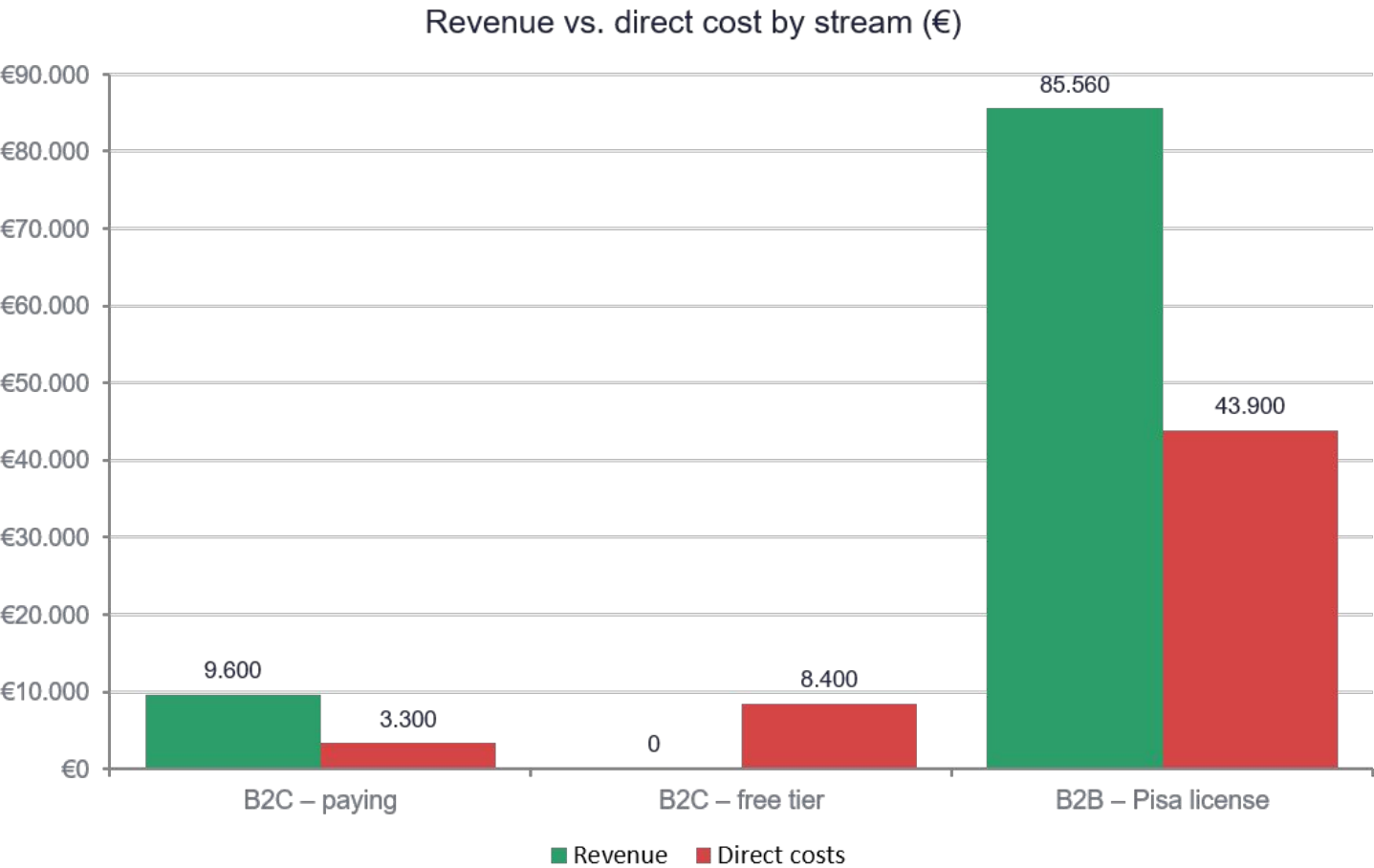
SOM

28,520 covered students (Pisa)

~€85,560/yr

Bottom-up sizing, B2C: Europe tertiary students (19.1M) → Italy target fields (62% of 2.3M) → first-year segment (330K students per year) → realistic conversion funnel (30% reach · 25% sign-up · 4% paid).

Year-1 picture: B2B profit funds B2C data acquisition



Stream	Margin
B2C – paying users	+ €6,300
B2C – free tier	– €8,400
B2B – Pisa license	+ €41,700
Total Year 1	+ €39,500

Break-even: ~4 Pisa-sized licenses on today's sales mix

50%

Overall CM ratio
(weighted, B2C 66% + B2B 49%)

€188.4K

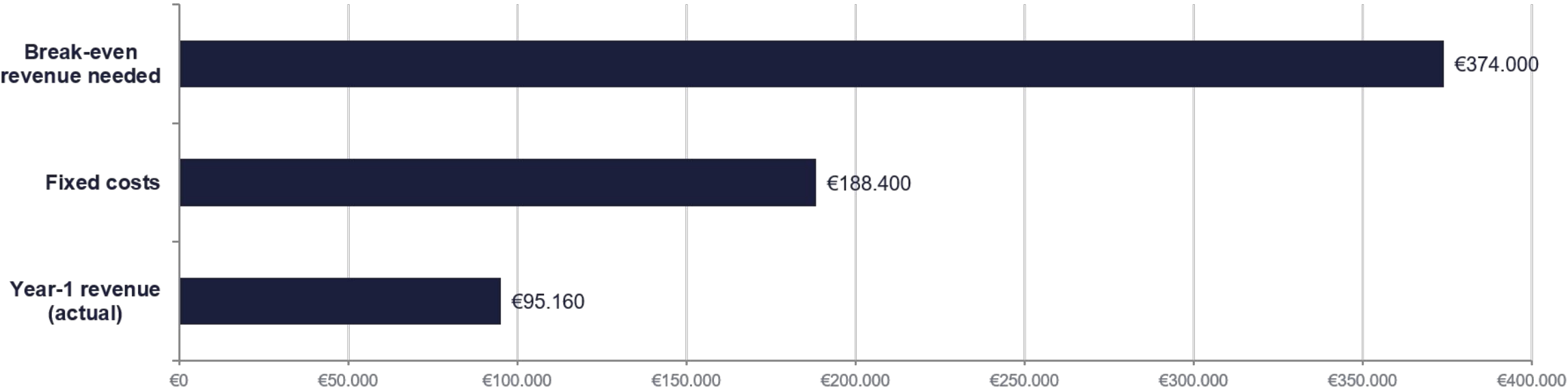
Total fixed costs

€374K

Break-even revenue
needed per year

≈4

Pisa-sized licenses
(≈3 more than Year 1)



What the numbers tell us

1

The market is large

Even restricting to Italian target fields: ~€22M/yr B2C and ~€4M/yr B2B. The European TAM is far larger (much broader market opportunity).

2

A Year-1 loss is normal

In early-stage startups costs grow faster than revenue; profitability stabilizes later. Year 1's goal is data and users, not profit.

3

Break-even is reachable

Closing the gap needs ~3 more institutional licenses beyond Pisa: this is the plan for Year 2 (while the B2C base keeps growing).

4

Speed is the bottleneck

Not market size. B2C converts only in seasonal pre-exam windows; B2B is a slow, budget-constrained sales cycle.

Bottom line

The opportunity justifies the investment: even a single institutional license already makes it worth pursuing and full break-even is a realistic Year-2 goal (~3 more licenses, B2C base keeps growing).

Who we build for



The Anxious Student

B2C buyer & user

GOAL

A safe space to fail, objective feedback and repeated practice to manage pre-exam panic.



The First-Timer

B2C buyer & user

GOAL

Understand what examiners expect, experience the format of an oral exam and find weak topics while there's still time to fix them.



The University Decision-Maker

B2B buyer, not user

GOAL

Improve pass rates & retention with a scalable tool, without taking on regulatory risk. Improve the quality of services for the students provided by the university.

Cheap and fast, then slow and expensive



Persevere · Pivot · Stop → decisions rest on evidence (fixed thresholds).

Stop conditions:

*below 30% of interviewed students confirm the problem → demand hypothesis is wrong, no iteration fixes it
the pilot shows low usage rate → B2B value proposition collapses*

Credible causes AlmReady could fail

1

Regulatory prohibition

AI Act exemption narrows (the 'no emotion inference' or 'non-certifying' line moves)

2

Increase of competition

Yoodli/Poised ship an 'exam mode'

3

B2C adoption failure

Reach/sign-up/conversion under threshold

4

B2B sales failure

2nd/3rd university deals stall

5

Technological failure

Latency or off-topic questions break trust

6

Cost-structure risk

Voice-API price increase; free-tier/university usage growing several times beyond the estimations

We build the escape route before we need it

Regulatory

Fallback audio-only mode that drops video/body-language analysis but keeps non-emotion delivery metrics

Competition

Grow the proprietary dataset aggressively from day one; stay narrowly focused on exam anxiety

Cost-structure

Keep a cheaper TTS provider ready as a fallback; cap free-tier sessions per user

AlmReady

Turning exam-day panic into practiced confidence... telling to yourself “I AM READY!”.

Thank you

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